

The Lorentz endpoint problem for elliptic canceling operators

Exact positive answer in later literature

Literature-status note for `fa_banach_001`

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Status: literature already answered. Stolyarov's endpoint Hardy–Littlewood–Sobolev theorem gives the estimate in Spector's Open Problem 7.3 for every elliptic canceling operator. This is not an original result of the run.

Original question

Spector asks in Open Problem 7.3, page 19 of arXiv:1903.09292 [1]: if $d \geq 3$ and the first-order homogeneous constant-coefficient operator

$$A(D) : C_c^\infty(\mathbb{R}^d; V) \longrightarrow C_c^\infty(\mathbb{R}^d; E)$$

is elliptic and canceling, must there be a constant C such that

$$\|u\|_{L^{d/(d-1),1}(\mathbb{R}^d; V)} \leq C \|A(D)u\|_{L^1(\mathbb{R}^d; E)} \quad (u \in C_c^\infty(\mathbb{R}^d; V))?$$

The source identifies this as, in particular, Van Schaftingen's Open Problem 8.3.

Exact later answer

For an elliptic homogeneous operator A of order m , Stolyarov associates the smooth Fourier-restriction field

$$\Omega(\zeta) = \operatorname{Im} \mathbb{A}(\zeta), \quad \zeta \in S^{d-1}.$$

Canceling means precisely $\bigcap_{\zeta \in S^{d-1}} \Omega(\zeta) = \{0\}$. Corollary 1.7 of arXiv:2010.05297 [2], obtained from the stronger Theorem 1.1, proves the optimal Lorentz endpoint embedding. In the elliptic differential-operator case its stated consequence is

$$\|\nabla^{m-1}u\|_{L^{d/(d-1),1}} \lesssim \|A(D)u\|_{L^1}.$$

Taking $m = 1$ makes $\nabla^{m-1}u = u$ and gives exactly Spector's requested inequality (indeed in every $d \geq 2$ where the formulation applies).

The supporting paper also states immediately after Corollary 1.7 that, for the rational field Ω coming from an elliptic differential operator and $\alpha = 1$, the corollary solves Van Schaftingen's Open Problem 8.3. Thus the problem-number cross-reference and the formula both match the source question.

References

- [1] D. Spector, *New Directions in Harmonic Analysis on L^1* , arXiv:1903.09292 (2019), Open Problem 7.3.
- [2] D. M. Stolyarov, *Hardy–Littlewood–Sobolev inequality for $p = 1$* , arXiv:2010.05297v3 (2021), Theorem 1.1 and Corollary 1.7; Sb. Math. 213 (2022), 844–889.